# Escaping Engine Lock-In: Open Source Tech Stacks

# About Alex

- IT & gamedev educator @ AIT since 2016
- Freelance & solo dev @ BigfootDS since 2012
- Began with graphic design & 3D animation, now 99% focused on programming
- Building open-source tools since 2020







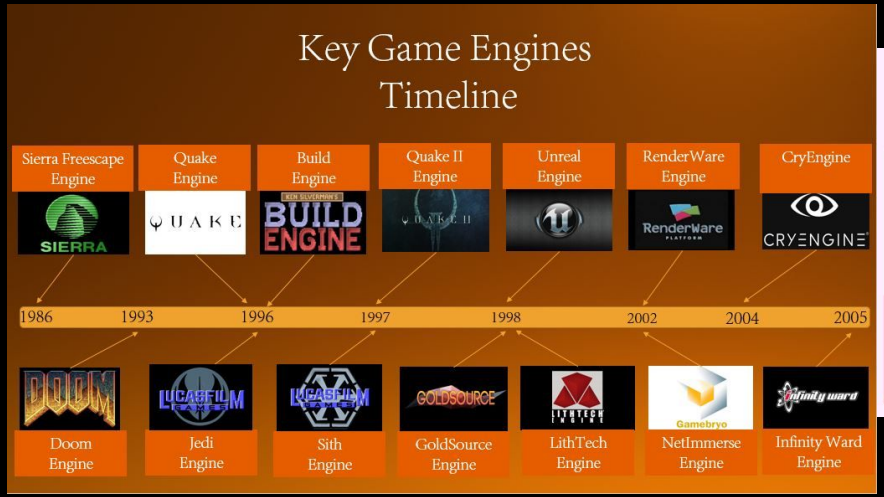


# This Talk

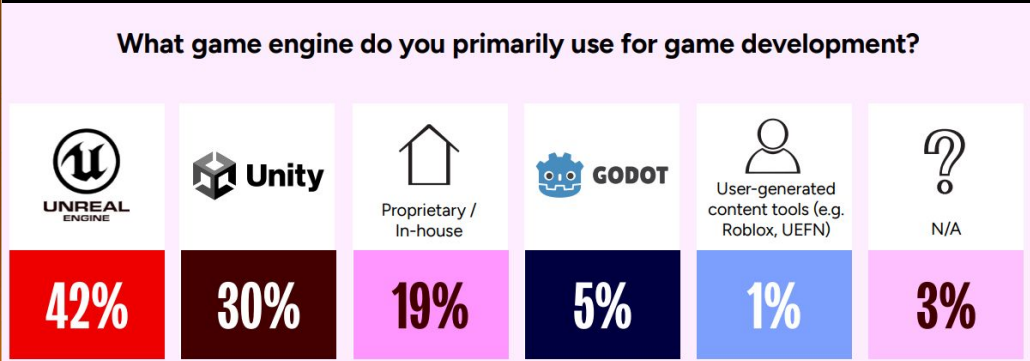
- Gamedev Projects & Typical Dependencies
- Breaking Gamedev Projects Down
- Tools & Solutions

# Why We Went To Engines In The First Place

- Studios don't have to "reinvent the wheel", saving time & resources
- Staff can find work more-easily and "hit the ground running" more quickly



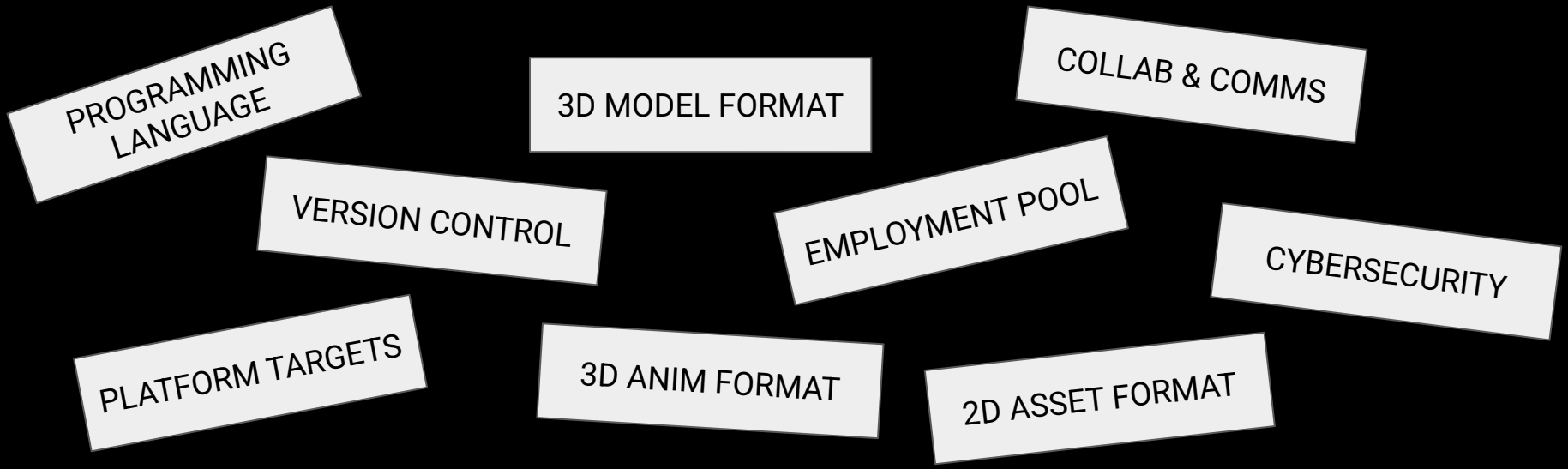

Key Game Engine Timeline by Luca Jorgenson, 2017




2026 STATE OF THE GAME INDUSTRY report from GDC, page 28, 2026

# What is Engine Lock-In?

Choosing an engine creates dependencies that can be expensive to replace.




You're "locked in" when it's too expensive to change your dependencies.

# What is Engine Lock-In?

the Guardian

**Game developers furious as Unity Engine announces new fees**

Unity announces a ‘Runtime Fee’, which will charge developers each time a game using the engine is downloaded




Article from 2023:
https://www.theguardian.com/games/2023/sep/12/unity-engine-fees-backlash-response

GamesIndustry.biz

**Tencent paid $330m for 40% share in Epic Games**

Chinese internet company Tencent paid $330 million for a 40 per cent stake in Unreal Engine maker Epic Games, according...



Article from 2013:
https://www.gamesindustry.biz/tencent-paid-usd330m-for-48-percent-share-in-epic-games

# What is Engine Lock-In? Not Just in Gamedev!

PetaPixel

## It's Cheaper to Fly to the US to Buy Adobe CS6 Than to Buy it In Australia



Here's a crazy fact that's making the rounds on the Internet: if you live in Australia, it's currently cheaper for you to fly to the US and...

14 Feb 2013

Macworld

## Adobe launches Creative Suite 6 alongside new Creative Cloud subscription service




The suspense is over. Adobe CS6 Master Collection Adobe has launched the much-anticipated sixth version of its Creative Suite—14 separate...

22 Apr 2012



# How Open Source Fits In

- Save money
- Digital independence & sovereignty
- Own your work & assets
- Own your workflows & processes



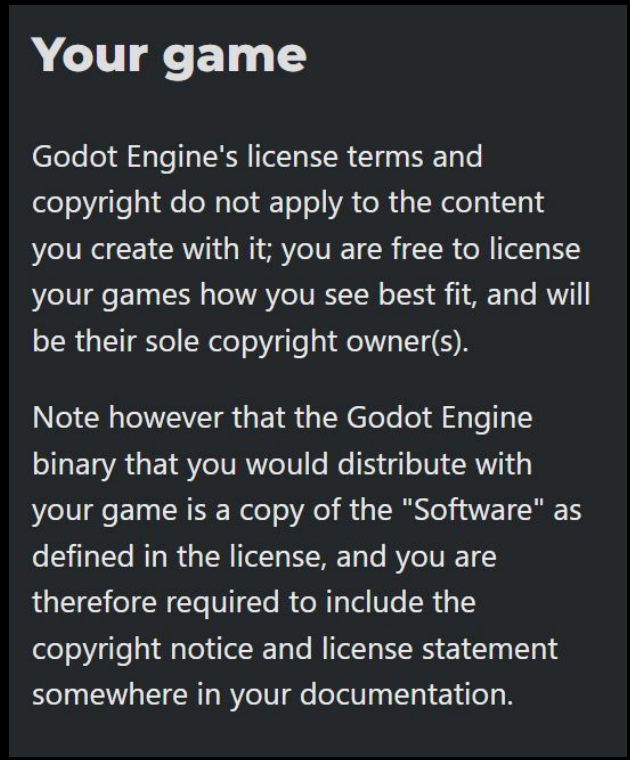
# How Open Source Fits In

Your game is your content.

Your tech stack is someone else's content.

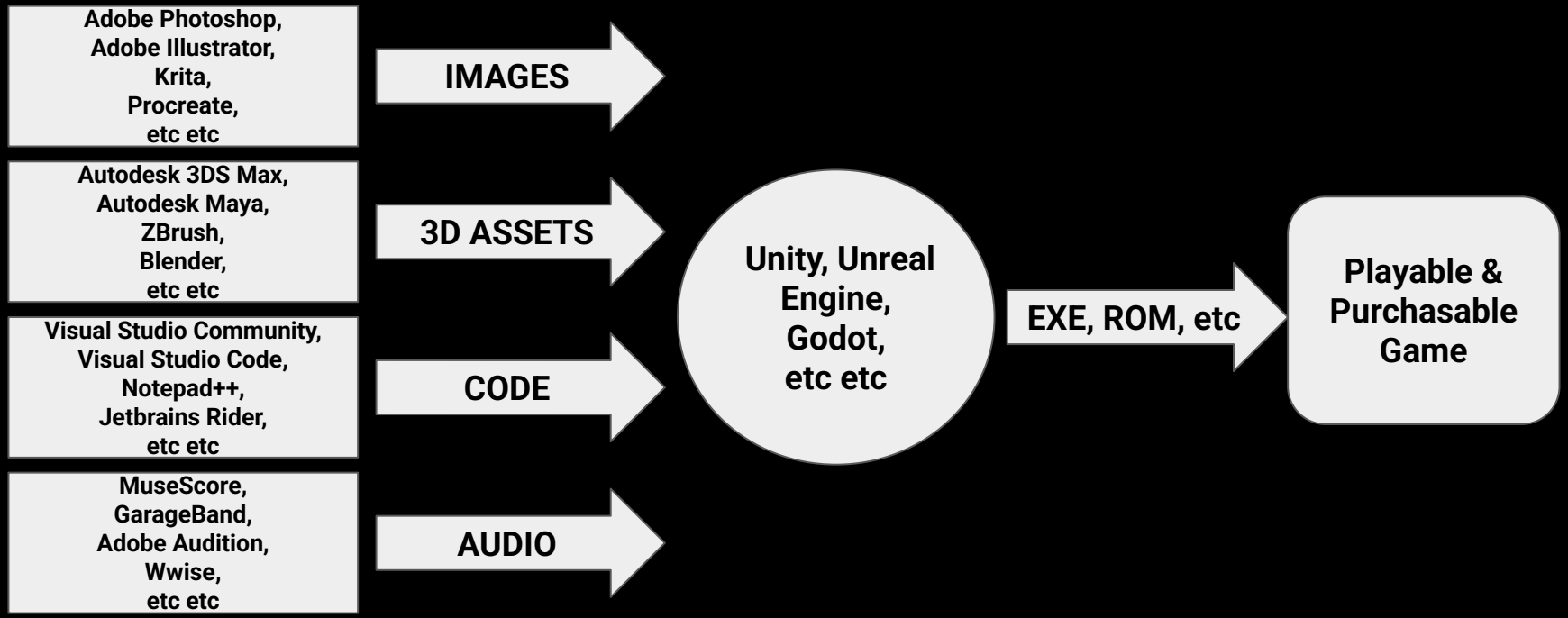
Find a way to make your tech stack your content!

This is easier to do when you break gamedev down into its component parts!

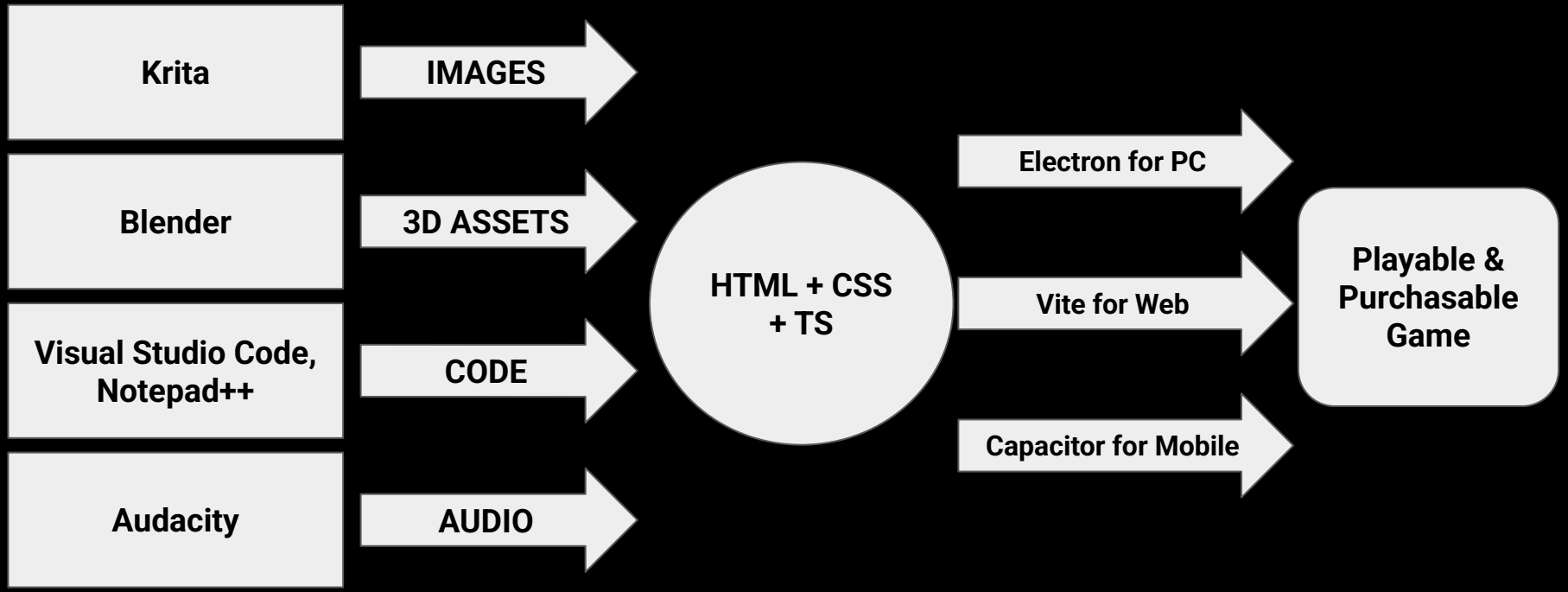



https://godotengine.org/license/

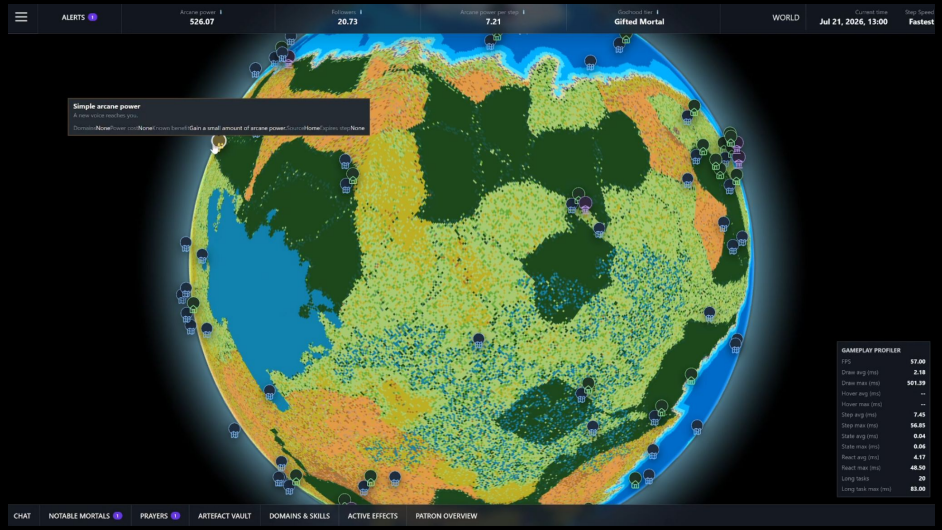
# How Open Source Fits In



# How Open Source Fits In - My Bubble



# How Open Source Fits In - My Bubble





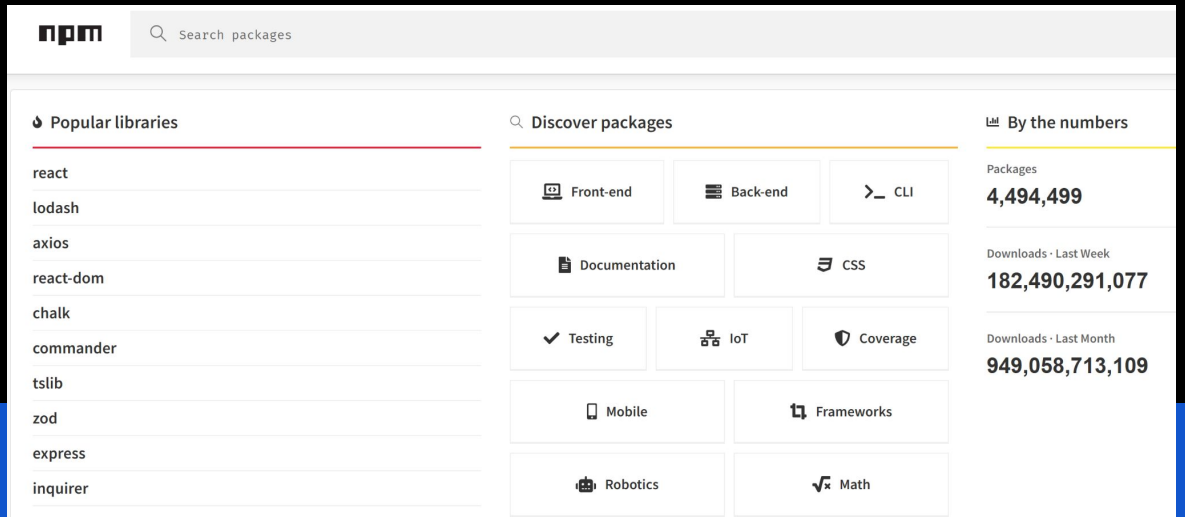

# How Open Source Works In Business

1. How to find & evaluate open source tools for your teams
2. How to keep projects safe and maintainable for the long-term
3. Cool tricks & bespoke things that we can do with open source tools

# How Open Source Works In Business

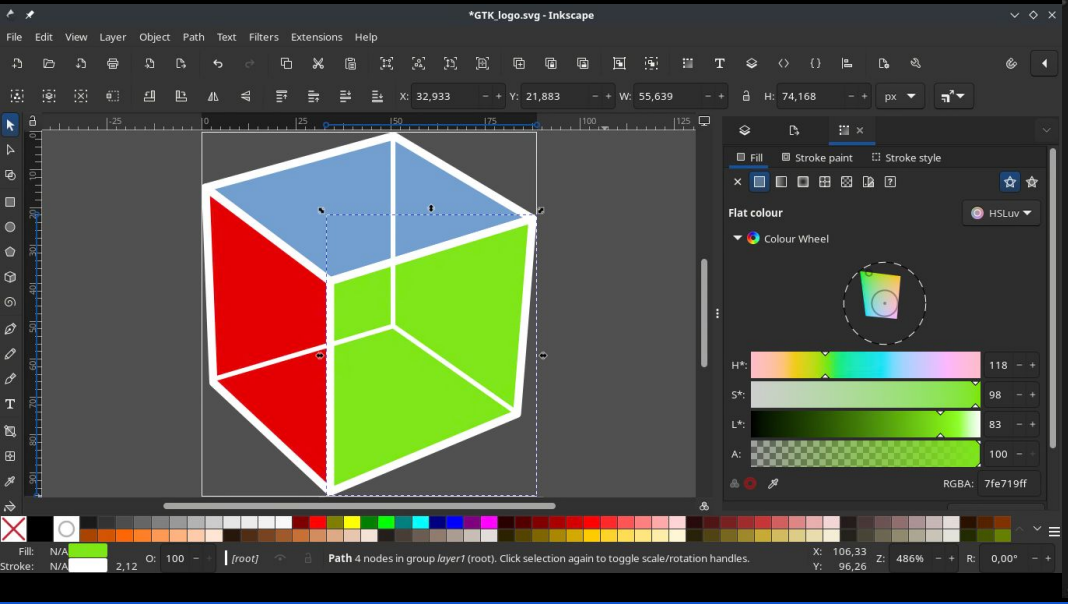
Don’t reinvent the wheel.

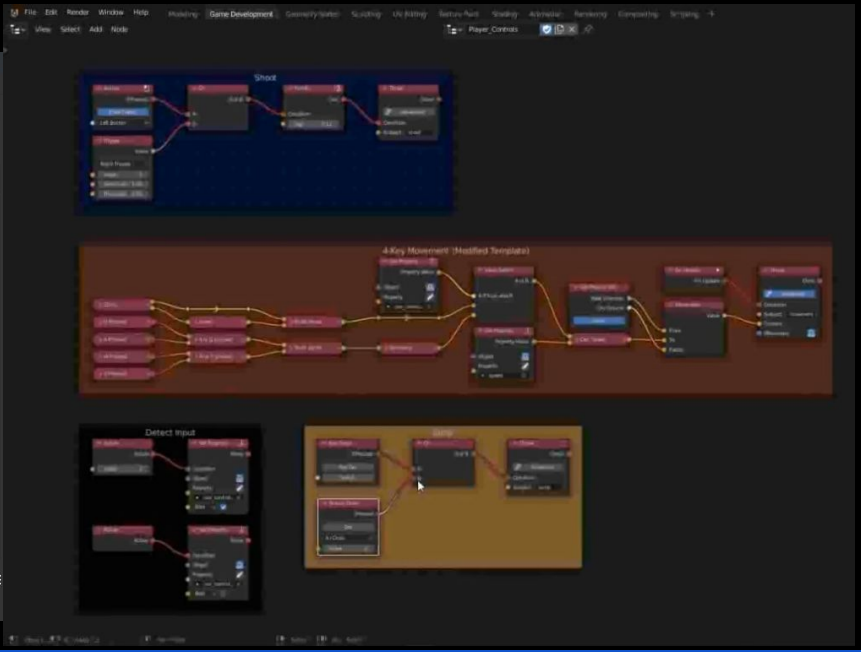
- Videogames are software. Look at what other types of software do!
- Search for features, not the whole game.



# How Open Source Works In Business

Let your skilled specialists make their outputs using their field-specific tools!



# How Open Source Works In Business

How to keep projects safe and maintainable for the long-term - ask questions!

- Does it support your development environment?
- Does it support your game's target platforms?
- Does it have good documentation and/or community for support?
- Does it work well for devs in my community?
- Does it have its own pain points if we have to replace it later?

# How Open Source Works In Business

How to keep projects safe and maintainable for the long-term

Switching to custom internal tools (not FOSS)

**Ludeon Studios – Announcing the Waylith Engine and Update 1.7**

**gatsu85vt** • 16h ago

They didn't rewrite the game from scratch, and they definitely didn't fork Unity. The secret is that RimWorld was never built like a typical Unity game in the first place. Ludeon never relied on Unity's GameObject hierarchy, scene graph, or components. From day one, the entire simulation—pawn AI, pathfinding, temperature, combat—has just been decoupled C# running directly in memory.

Unity was basically just a rendering shell: handling the window, input, audio, and batch-drawing 2D sprites through raw draw calls. Switching to Waylith only required swapping out that outer presentation layer for their own custom backend and modern .NET runtime, leaving the actual simulation untouched. To keep the modding ecosystem alive, they use shim libraries that intercept and translate legacy Unity API calls. It sounds crazy from the outside, but it's just the huge payoff of keeping game logic strictly separated from engine rendering.



# How Open Source Works In Business

How to keep projects safe and maintainable for the long-term

Switching to FOSS

MegaCrit's Godot transition experience is detailed here:
https://caseyyano.com/on-evaluating-godot-b35ea86e8cf4

Godot Engine

**Slay the Spire 2 – Godot Engine**

The iconic roguelike deckbuilder returns. Craft a unique deck, encounter bizarre creatures, and discover relics of immense power in Slay the Spire 2!




MEGA CRIT

Slay the Spire

The Mega Crit team has been hard at work these past 2+ years on a new game. But unlike with Slay the Spire, the engine we have been developing it in is Unity.

The retroactive pricing structure of Runtime Fees is not only harmful in a myriad of ways to developers--especially indies--it is also a violation of trust. We believe Unity is fully aware of this, seeing as they have gone so far as to remove their TOS from GitHub.

Despite the immense amount of time and effort our team has already poured into development on our new title, we will be migrating to a new engine unless the changes are completely reverted and TOS protections are put in place.

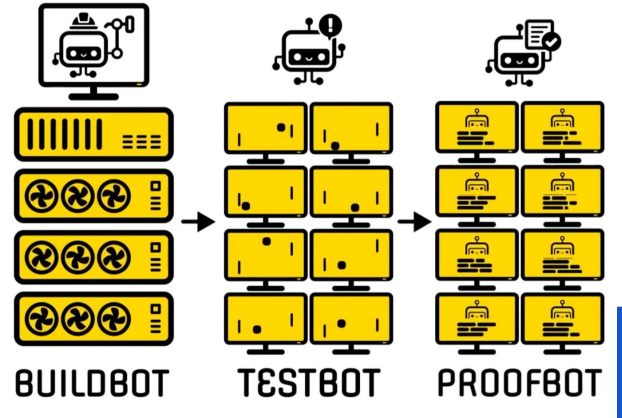
**We have never made a public statement before. That is how badly you fucked up.**

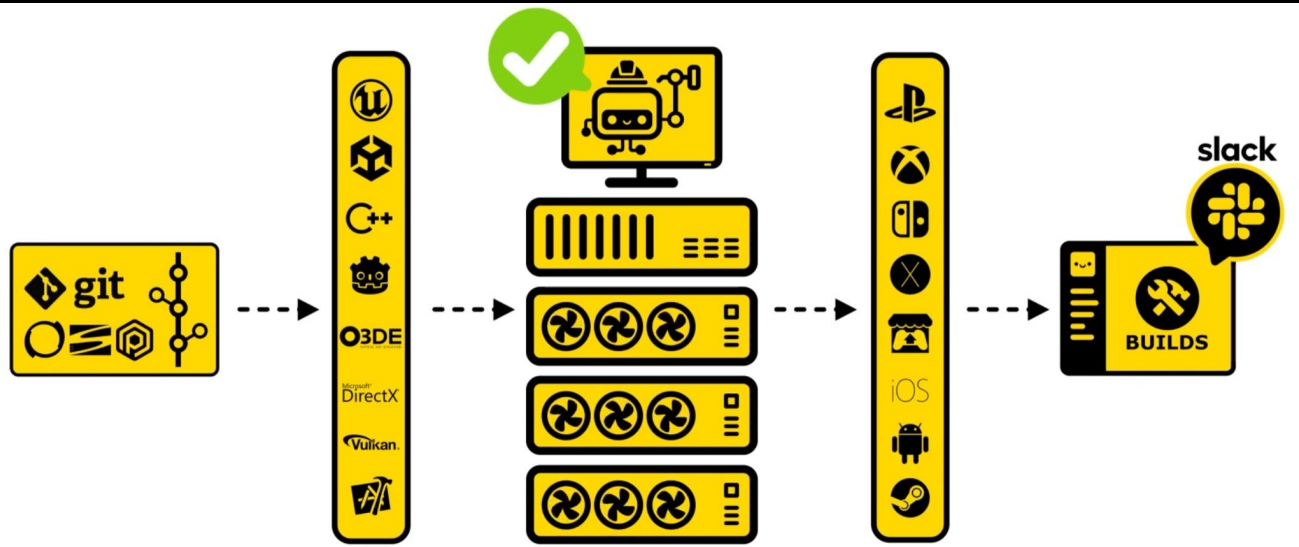
BigfootDS

# How Open Source Works In Business

Cool tricks & bespoke things that we can do with open source tools:

- Automation!
- Internal tools!
- New products!




All About Mighty Build and Test, 18th August 2023, https://www.youtube.com/watch?v=gVcT2nz4V0U

# Open Source Gamedev Whirlwind Tour

The common stuff:

- Art via Krita and Inkscape
- Audio via Audacity and royalty-free soundpacks
- Code via Visual Studio Code
- Programming via HTML, CSS, JavaScript
- Compiled via ElectronJS and CapacitorJS

PC & smartphone multiplatform game
from a free tech stack



# Open Source Gamedev Whirlwind Tour

The problem-solving stuff:

- “Waiting for the game to build sucks”
- “Testing is annoying”
- “Uploading to Steam is so tedious”

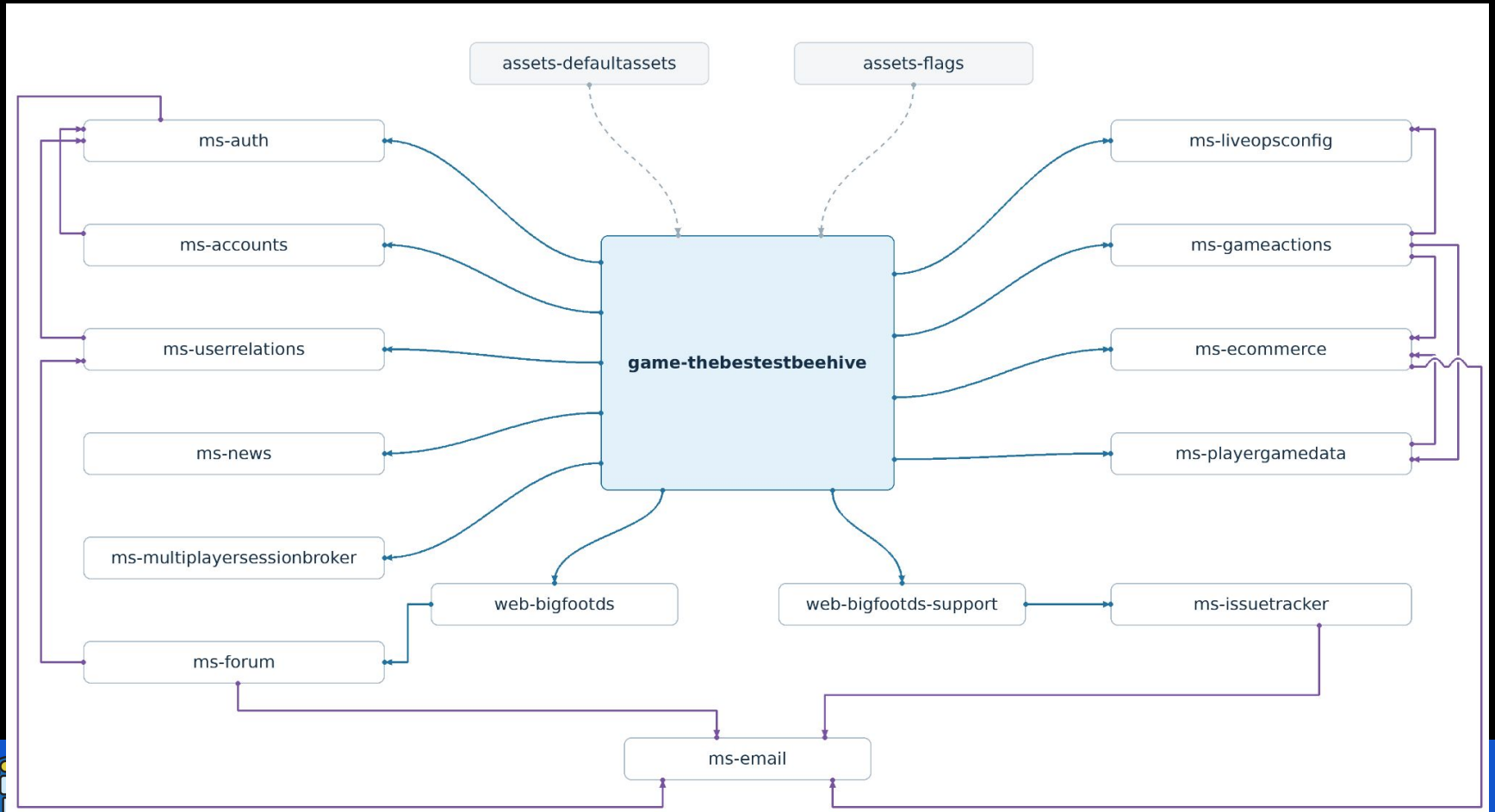





# Open Source Gamedev Whirlwind Tour

- CI/CD
  - Automated testing
  - Automated building
  - Automated publishing
- CI/CD/CC
  - Automated credits lists
  - Automated version numbering
  - SBOMs!
- Design Tools
  - Custom TCG card & set product designer
  - Data simulations for data-heavy games
  - Asset packers
  - Living design docs with engine integration
  - Literal typical software dev
- Game Infrastructure
  - LiveOps
  - ChatOps
  - Literal typical webdev

# Open Source Gamedev Whirlwind Tour



# Solution Shopping List - Part 1

| ASPECT | FREE or OPEN-SOURCE | TYPICAL TOOLS |
|---|---|---|
| Game Engine | Godot, UPBGE, a DIY framework | Unity, Unreal Engine |
| Programming | Visual Studio Code, VSCodium | Visual Studio Community |
| 2D Illustration & Image Editing | Krita, GIMP, Darktable | Adobe Photoshop, Procreate, Adobe Lightroom |
| Vector Art | Inkscape | Adobe Illustrator |
| Pixel Art & Sprite Animation | Pixelorama | Aseprite |
| UI/UX Design & Prototyping | Penpot, Pencil | Figma, Sketch, Balsamiq |
| 3D Modelling | Blender | Autodesk 3DS Max, Autodesk Maya |
| Digital Sculpting | Blender | ZBrush |

# Solution Shopping List - Part 2

| ASPECT | FREE or OPEN-SOURCE | TYPICAL TOOLS |
|---|---|---|
| Texture Painting | Blender, ArmorPaint | Substance 3D Painter, Mari |
| Procedural Materials | Material Maker | Substance 3D Designer |
| Procedural Modelling | Blender Geometry Nodes | Houdini |
| 2D Animation | OpenToonz, Krita, Synfig | Toon Boom Harmony, TVPaint, Moho |
| 3D Animation & Rigging | Blender | Autodesk Maya, Autodesk MotionBuilder |
| Narrative Design | Ink, Twine | articy:draft |
| Music Composition & Production | LMMS, Ardour | FL Studio, Cubase, REAPER |
| Sound Editing & Recording | Audacity, Ardour | Adobe Audition, REAPER, Pro Tools |

# Solution Shopping List - Part 3

| ASPECT | FREE or OPEN-SOURCE | TYPICAL TOOLS |
|---|---|---|
| Video Editing | Kdenlive, Shotcut | Adobe Premiere Pro, DaVinci Resolve |
| Gameplay Capture & Live Streaming | OBS Studio | Bandicam, Camtasia |
| Version Control & Large Assets | Git + Git LFS | Perforce P4 |
| Version Control Hosting | Forgejo | GitHub, GitLab, Bitbucket |
| Automation | Forgejo Actions, Jenkins | GitHub Actions, TeamCity |
| Project Management & Issue Tracking | Forgejo Projects, OpenProject Community | GitHub Projects, Jira, Linear, Trello |
| Documentation & Design Wikis | Forgejo Wiki, BookStack, DIY website or wiki | Confluence, Notion |
| QA & Test-Case Management | Kiwi TCMS | TestRail |

# Solution Shopping List - Part 4

| ASPECT | FREE or OPEN-SOURCE | TYPICAL TOOLS |
|---|---|---|
| Multiplayer & Game Back-ends | Nakama, DIY back-ends | PlayFab, Unity Gaming Services |
| Websites | WordPress, Astro, DIY front-ends | Squarespace, Webflow |
| Custom APIs & Back-ends for Data | DIY back-ends | Google Firebase |
| Feature Flags & Rollout Controls | Unleash Open Source, DIY back-ends | LaunchDarkly |
| Localisation Management | Weblate | Crowdin, Phrase |
| Graphics Debugging & Profiling | RenderDoc, Tracy | PIX, NVIDIA Nsight |
| Social Media Scheduling | Postiz | Buffer, Hootsuite |
| Player Support & Ticketing | Zammad Open Source | Zendesk, Freshdesk |

# Solution Shopping List - Part 123456789…

You need to explore open source!

Share your experiences, or better: improve open source tools to meet your needs!



# Summary

- We don’t have to use someone else’s built-for-them solution for our goals
- Games are made up of work from multiple industries - we don’t always need “built for gamedev” solutions to do not-strictly-gamedev work!
- There are a lot of solutions out there. Exploration helps everybody grow!

You don’t need to rewrite your whole game project to try open source tools!
Break your project into smaller parts.
Give ANYTHING a go!

# Contact Me

**Alex Stormwood**

alex.stormwood@bigfootds.com

alexstormwood.com

bigfootds.com

github.com/AlexStormwood

linkedin.com/in/alexstormwood

These slides and more info:

